

# Quantitative Research Methods

## GLMs and Splines

Advanced module A4 — optional self-study

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# Where this module fits

This is an **optional advanced module**: nothing in it is required for the exam, and everything in it deepens material you have already met.

| Course chapter             | What this module adds                                    |
|----------------------------|----------------------------------------------------------|
| Ch. 3 — Linear regression  | ... becomes the <i>Gaussian corner</i> of one big family |
| Ch. 4 — Classification     | ... logistic and Poisson regression re-derived as GLMs   |
| Ch. 5 — Resampling         | ... CV and GCV choose the spline penalty $\lambda$       |
| Ch. 6 — Model selection    | ... deviance, likelihood-ratio tests and AIC for GLMs    |
| ► Ch. 7 — Splines and GAMs | ... <b>penalized splines, and GAMs for count data</b>    |

## One sentence

Chapter 4's likelihood view and Chapter 7's smooth curves are two halves of the same machine — this module assembles it: **exponential family** → **GLM** → **penalized spline** → **GAM for counts**, all on one real dataset (Bikeshare).

# Contents

- 1 Counts Break the LM
- 2 Exponential Family
- 3 GLM Anatomy
- 4 Inference
- 5 Overdispersion
- 6 Penalized Splines
- 7 GAMs for Counts
- 8 Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

# Notation in this module

| Symbol                           | Meaning                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------|
| $Y_i, y_i$                       | response (here: hourly count of bikers); its observed value                       |
| $\mu_i = E[Y_i]$                 | mean of the response for observation $i$                                          |
| $\theta, \phi$                   | canonical (natural) parameter; dispersion parameter                               |
| $b(\theta), a(\phi), c(y, \phi)$ | the three building blocks of an exponential-family density                        |
| $V(\mu)$                         | variance function: $\text{Var}(Y) = a(\phi) V(\mu)$                               |
| $\eta_i = x_i^\top \beta$        | linear predictor                                                                  |
| $g(\cdot)$                       | link function, $g(\mu_i) = \eta_i$ ; canonical link $g = (b')^{-1}$               |
| $D, D_0$                         | residual deviance; null deviance (intercept-only model)                           |
| $\hat{\phi}$                     | Pearson estimate of the dispersion (quasi-Poisson)                                |
| $\alpha$                         | negative-binomial overdispersion: $\text{Var} = \mu + \alpha\mu^2$                |
| $B_j(x), \beta_j$                | $j$ -th B-spline basis function; its coefficient                                  |
| $\lambda, S$                     | smoothing penalty weight; penalty matrix, $\int f''(t)^2 dt = \beta^\top S \beta$ |
| edf                              | effective degrees of freedom, $\text{tr}\{H(\lambda)\}$                           |
| $f_j(\cdot)$                     | smooth partial effect of predictor $j$ in a GAM                                   |

## Sources and references

### Primary textbook

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python*. Springer. — Sections 4.6 (Poisson regression) and Chapter 7 (splines, GAMs).

### Going deeper (optional)

McCullagh, P. & Nelder, J. (1989). *Generalized Linear Models*, 2nd ed., Chapman & Hall.  
Wood, S. (2017). *Generalized Additive Models: An Introduction with R*, 2nd ed., CRC Press.  
Eilers, P. & Marx, B. (1996). Flexible smoothing with B-splines and penalties. *Statistical Science*, 11(2).

# Learning objectives

By the end of this module you will be able to:

- **Explain** the exponential family and derive a member's mean and variance from  $b(\theta)$ .
- **Specify** a GLM — random component, linear predictor, link — for a given data type.
- **Interpret** Poisson coefficients as rate ratios, and **use** offsets for exposure.
- **Test** nested GLMs with deviance and likelihood-ratio tests, and **compare** models with AIC.
- **Diagnose** overdispersion and **choose** between quasi-Poisson and negative binomial.
- **Tune** a penalized spline via GCV or CV and **read** its effective degrees of freedom.
- **Fit** and **interpret** a penalized-spline GAM for counts with `statsmodels`.

# Roadmap of this module

## From a broken linear model to a GAM for counts

- 1 Why the linear model fails on Bikeshare counts.
- 2 The exponential family: one density form behind Chs. 3 and 4.
- 3 GLM anatomy: random component + linear predictor + link; Poisson regression, offsets.
- 4 Inference: deviance, likelihood-ratio tests, AIC.
- 5 Overdispersion: quasi-Poisson and the negative binomial.
- 6 Splines deepened: B-spline bases, roughness penalties, effective df, GCV.
- 7 The meeting point: penalized-spline GAMs for counts.

# Where this module is used in industry

| Sector             | Concrete application                                                                        | Which decision it drives             |
|--------------------|---------------------------------------------------------------------------------------------|--------------------------------------|
| Insurance          | Claim <i>frequency</i> as a Poisson/NB GLM with exposure offsets; age curves as GAM smooths | The premium each tariff cell pays    |
| Retail, e-commerce | Daily demand counts per SKU and store, weather and calendar smooths                         | Ordering, staffing and markdowns     |
| Energy             | Load forecasting with GAMs: smooth effects of temperature and time of day                   | How much generation to schedule      |
| Public health      | Case and admission counts vs exposure population                                            | Staffing, stockpiles, early warnings |
| Banking            | Default (yes/no) as a logistic GLM — the same machinery                                     | Approve or decline; capital held     |
| Mobility platforms | Hourly ride counts vs hour, weather, events (this module's dataset)                         | Fleet rebalancing and surge staffing |

## In industry — The common thread

Whenever the response is a **count** or a **rate**, industry reaches for exactly this stack: GLM for interpretability and valid uncertainty, smooth terms for the shapes no one wants to guess, offsets for unequal exposure.

# Industry case in depth: pricing motor insurance

## In industry — Claim frequency at a motor insurer

A pricing team models the **number of claims** per policy-year for millions of policies. Policies are observed for unequal durations — three months here, a full year there — so the model is a Poisson GLM for the claim *rate* with  $\log(\text{exposure})$  as an **offset**. Tariff factors (region, vehicle class, bonus–malus) enter as dummies; **driver age** enters as a penalized-spline smooth, because its effect is famously U-shaped and no polynomial degree is defensible to a regulator.

## Why the family matters commercially

The premium is roughly frequency  $\times$  severity. If the frequency model ignores **overdispersion**, its standard errors are far too small: tariff cells look “significantly” different when they are not, and the tariff fragments into noise. Actuarial practice therefore fits quasi-Poisson or negative binomial by default.

## Takeaway

Every ingredient of this module — log link, rate ratios, offsets, overdispersion, spline smooths, edf — appears in one production pricing model.

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# The Bikeshare data: 8 645 hours of counts

Hourly bike rentals in Washington DC (ISLP): response bikers, the number of riders in that hour.

| Variable   | Type                   | Range / levels                                 | Role                     |
|------------|------------------------|------------------------------------------------|--------------------------|
| bikers     | count                  | 1 to 651                                       | response                 |
| hr         | hour of day            | 0–23                                           | 23 dummies (or a smooth) |
| temp       | normalised temperature | 0.02–1 ( $1 \approx 41^\circ\text{C}$ )        | numeric                  |
| workingday | binary                 | 0/1                                            | dummy                    |
| weathersit | categorical            | clear / cloudy-misty / light rain / heavy rain | 3 dummies                |

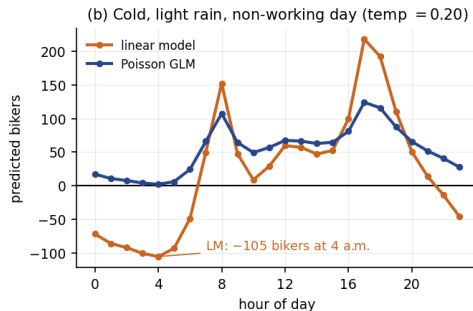
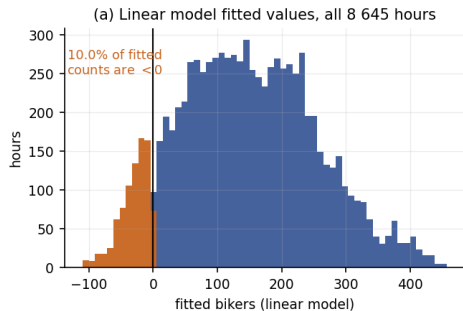
## The response at a glance

$n = 8\,645$  hours; mean  $\bar{y} = 143.8$  bikers, variance  $s^2 = 17\,902$  — the variance is about  $124\times$  the mean. Weather levels are very unbalanced: clear 5 645, cloudy/misty 2 218, light rain/snow 781, heavy rain/snow a single hour.

## Takeaway

A count response lives on  $\{0, 1, 2, \dots\}$  and its spread grows with its level. Keep both facts in mind — the linear model respects neither.

# Fit the Chapter-3 linear model ... and watch it fail



## Takeaway

OLS on  $\text{bikers} \sim \text{hr} + \text{temp} + \text{workingday} + \text{weathersit}$ : **10.0%** of fitted values (861 of 8 645 hours) are *negative*; on a cold rainy night it predicts -105 riders at 4 a.m. — the Poisson GLM (right) predicts 2.2: small, positive, sensible.

## Two failures, one diagnosis

### What exactly is wrong with the Gaussian linear model here?

Model under indictment:  $Y = x^\top \beta + \varepsilon, \varepsilon \sim N(0, \sigma^2)$ .

- ❶ **Wrong support.** A Gaussian mean can be negative; a count cannot. The 10.0% negative predictions are the model doing what it was told.
- ❷ **Wrong variance.** Constant  $\sigma^2$  — but quiet 4 a.m. hours vary by a few riders, busy 5 p.m. hours by hundreds: *variance grows with the mean*.

### What we need instead

A model whose **mean respects the data type**, whose **variance is tied to the mean**, and which keeps a **linear predictor**  $x^\top \beta$  we know how to estimate and interpret. That is precisely a *generalized linear model*.

### In industry — Mobility platforms: why negative forecasts hurt

A fleet-rebalancing optimiser fed “−105 rides expected” will happily truck bikes *away* from a station that will see riders at dawn.

## Exercise A4.1 — Choosing the random component [Concept]

### Exercise

- 1 Name the **two** distinct failures of the Gaussian linear model on hourly bike counts, and say which *observable symptom* each produces.
- 2 For each response below, propose a distribution and a link that respect the data type: (a) number of insurance claims per policy-year; (b) whether a loan defaults; (c) a person's log-wage.
- 3 A colleague proposes fixing the negative predictions by modelling  $\log(y + 1)$  with OLS. Give one reason this is *not* equivalent to a Poisson GLM.

## Solution — Exercise A4.1

### Solution

**Method.** Match the *support* and the *mean–variance relation* of the response to a distribution; the link keeps the mean inside the support.

**Part 1.** (i) *Wrong support*: fitted means can be negative — symptom: 10.0% negative predictions on Bikeshare. (ii) *Wrong variance*: constant  $\sigma^2$  — symptom: residual spread visibly grows with the fitted level (a funnel in the residual plot).

**Part 2.** (a) counts  $\Rightarrow$  Poisson with log link (offset for exposure); (b) binary  $\Rightarrow$  Bernoulli with logit link — exactly Chapter 4’s logistic regression; (c) continuous, roughly symmetric  $\Rightarrow$  Gaussian with identity link — Chapter 3.

**Part 3.** OLS on  $\log(y + 1)$  models  $E[\log(Y + 1)]$ , not  $\log E[Y]$ : by Jensen’s inequality these differ, so back-transformed predictions are biased for the mean count; the “+1” is arbitrary; and the implied variance is again constant on the log scale rather than Poisson-like. The GLM models  $\log E[Y]$  directly.

### Common mistake

“Transform  $y$ , then use OLS” changes *what is being estimated*. A GLM transforms the **mean**, not the **data**.

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# One density to rule them all

## Definition (exponential dispersion family)

A response  $Y$  belongs to the exponential family if its density (or pmf) can be written

$$f(y; \theta, \phi) = \exp \left\{ \frac{y \theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}.$$

### How to read this

- $\theta$  — the **canonical parameter**: the natural scale on which  $y$  enters the exponent linearly.
- $b(\theta)$  — the **cumulant function**: its derivatives hand us mean and variance for free.
- $a(\phi)$  — the **dispersion**: a scale factor ( $\phi = 1$  for Poisson and Bernoulli).
- $c(y, \phi)$  — normalisation without  $\theta$ : never touches the estimation of  $\beta$ .

### Takeaway

Gaussian (Ch. 3), Bernoulli (Ch. 4) and Poisson (Ch. 4.6) are all of this form — one estimation theory covers all three.

# The mean and the variance come from $b(\theta)$

## Rule (mean–variance identities)

$$E[Y] = b'(\theta), \quad \text{Var}(Y) = b''(\theta) a(\phi) = a(\phi) V(\mu).$$

## How to read this

- Differentiate the log-likelihood in  $\theta$ , take expectations, and both identities fall out of the score equations (full derivation in the **appendix**).
- $V(\mu) = b''(\theta)$  rewritten as a function of  $\mu$  is the **variance function**: it states *how the spread must move with the mean* — the very thing OLS got wrong on counts.
- $a(\phi)$  scales the whole variance but leaves its *shape*  $V(\mu)$  alone.

## Takeaway

Choosing a family = choosing a mean–variance relation. That is the modelling decision; everything else is shared machinery.

# Worked derivation: the Poisson is a member

## Rewrite the Poisson pmf into family form

$$P(Y = y) = \frac{e^{-\mu} \mu^y}{y!} = \exp\{y \log \mu - \mu - \log y!\}, \quad y = 0, 1, 2, \dots$$

Match against  $\exp\{(y\theta - b(\theta))/a(\phi) + c(y, \phi)\}$ :  $\theta = \log \mu$ ,  $b(\theta) = e^\theta$ ,  $a(\phi) = 1$ ,  $c(y, \phi) = -\log y!$ .  
 Cash in the identities:  $E[Y] = b'(\theta) = e^\theta = \mu$  ✓ and  $\text{Var}(Y) = b''(\theta) \cdot 1 = \mu$  — the famous **mean = variance** property.

## Sanity check with numbers

With  $\mu = 2$ :  $P(Y = 3) = e^{-2} 2^3 / 3! = 0.1804$ ; canonical parameter  $\theta = \log 2 = 0.693$  — the *log-mean* scale on which Poisson regression will be linear.

## Takeaway

The canonical parameter of the Poisson is  $\log \mu$  — the log link is not a convention, it is **built into the distribution**.

# The three members you already know

| Family    | $\theta(\mu)$            | $b(\theta)$          | $a(\phi)$  | $E[Y] = b'(\theta)$                 | $\text{Var}(Y) = b''(\theta)a(\phi)$ |
|-----------|--------------------------|----------------------|------------|-------------------------------------|--------------------------------------|
| Gaussian  | $\mu$                    | $\theta^2/2$         | $\sigma^2$ | $\theta = \mu$                      | $\sigma^2$                           |
| Bernoulli | $\log \frac{\mu}{1-\mu}$ | $\log(1 + e^\theta)$ | 1          | $\frac{e^\theta}{1+e^\theta} = \mu$ | $\mu(1 - \mu)$                       |
| Poisson   | $\log \mu$               | $e^\theta$           | 1          | $e^\theta = \mu$                    | $\mu$                                |

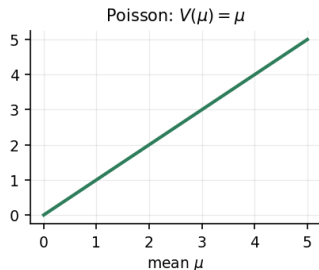
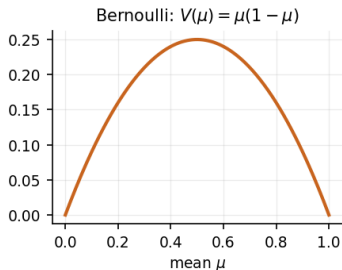
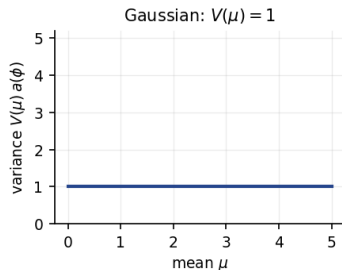
## Read each row the same way

Column 2 names the *natural scale* of the mean (identity, log-odds, log); the last column is the **variance function** — constant, quadratic in  $\mu$  with a maximum at  $\mu = \frac{1}{2}$ , and proportional to  $\mu$ , respectively.

## Takeaway

The Gaussian derivation is in the appendix; the Bernoulli one is **Exercise A4.2** — next slide.

# Each family fixes how spread follows the mean



## Takeaway

Gaussian: flat variance, whatever the mean. Bernoulli: variance peaks at 0.25 when  $\mu = 0.5$  and vanishes at the ends. Poisson: variance climbs one-for-one with the mean — at  $\mu = 5$  the variance *must* be 5. Data that violate the chosen line (as Bikeshare will, spectacularly) violate the family.

## Exercise A4.2 — The Bernoulli as a family member [Math]

### Exercise

Let  $Y \in \{0, 1\}$  with  $P(Y = 1) = \mu$ , so  $f(y; \mu) = \mu^y(1 - \mu)^{1-y}$ .

- 1 Rewrite  $f$  in exponential-family form and identify  $\theta(\mu)$ ,  $b(\theta)$  and  $a(\phi)$ .
- 2 Verify  $b'(\theta) = \mu$ .
- 3 Verify  $b''(\theta) = \mu(1 - \mu)$ , and state in one sentence what this variance function implies for observations with  $\mu$  near 0.5 versus near 0.99.

## Solution — Exercise A4.2

## Solution

**Method.** Take logs, isolate the coefficient of  $y$ , and match term by term.

**Part 1.**  $\log f = y \log \mu + (1 - y) \log(1 - \mu) = y \log \frac{\mu}{1 - \mu} + \log(1 - \mu)$ , hence

$$\theta = \log \frac{\mu}{1 - \mu} \text{ (the logit)}, \quad a(\phi) = 1, \quad c(y, \phi) = 0.$$

Inverting the logit gives  $\mu = e^\theta / (1 + e^\theta)$ , so  $\log(1 - \mu) = -\log(1 + e^\theta)$  and therefore  $b(\theta) = \log(1 + e^\theta)$ .

**Part 2.**  $b'(\theta) = \frac{e^\theta}{1 + e^\theta} = \mu$  ✓ — the sigmoid of Chapter 4.

**Part 3.**  $b''(\theta) = \frac{e^\theta}{(1 + e^\theta)^2} = \mu(1 - \mu)$  ✓. Near  $\mu = 0.5$  the variance is maximal (0.25): most informative about  $\beta$ ; near  $\mu = 0.99$  it is 0.0099 — nearly deterministic, little information.

## Common mistake

$b(\theta)$  must be written as a function of  $\theta$ , not of  $\mu$  — differentiating  $b = -\log(1 - \mu)$  with respect to  $\theta$  as if  $\mu$  were constant gives nonsense.

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# The three components of a GLM

## Rule (GLM specification)

1. **Random component:**  $Y_i$  exponential-family with mean  $\mu_i$ . 2. **Systematic component:** linear predictor

$\eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}$ . 3. **Link:**  $g(\mu_i) = \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$

## How to read this

- The link  $g$  maps the mean's natural range onto the whole real line, where a linear predictor can roam freely:  $\log : (0, \infty) \rightarrow \mathbb{R}$ ,  $\text{logit} : (0, 1) \rightarrow \mathbb{R}$ .
- The **canonical link** sets  $g = (b')^{-1}$ , i.e.  $\theta_i = \eta_i$ : identity for Gaussian, logit for Bernoulli, log for Poisson — concave log-likelihoods, simple score equations.
- Estimation is by **maximum likelihood**, via iteratively reweighted least squares (IRLS — sketch in the appendix).

## Takeaway

Chapter 3 = Gaussian + identity. Chapter 4 = Bernoulli + logit. Today's workhorse = Poisson + log. Same anatomy, three organs swapped.

# Logistic regression, re-derived in one line

## Chapter 4 falls out as a special case

Take the Bernoulli member ( $\theta = \log \frac{\mu}{1-\mu}$ ) and impose the canonical link  $\theta_i = \eta_i$ :

$$\log \frac{p(x_i)}{1 - p(x_i)} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} \iff p(x_i) = \frac{e^{x_i^\top \beta}}{1 + e^{x_i^\top \beta}}.$$

This *is* equation (4.6) of ISLP — and “ $e^{\beta_j}$  multiplies the odds” is the Bernoulli twin of the rate ratios two slides ahead.

## Takeaway

Nothing about logistic regression was ad hoc: sigmoid, log-odds reading, ML fitting — all forced by “Bernoulli + canonical link”.

## In industry — Credit scoring: the regulated GLM

Default models in banking are Bernoulli GLMs: every  $e^{\beta_j}$  has a documented odds-ratio meaning — a key reason GLMs survive in the deep-learning era.

# Poisson regression: the model for Bikeshare

## Rule (Poisson GLM with log link)

$$Y_i \sim \text{Poisson}(\mu_i), \quad \log \mu_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} \iff \mu_i = e^{\beta_0} e^{\beta_1 x_{i1}} \cdots e^{\beta_p x_{ip}}.$$

## The multiplicative reading

- $\mu_i > 0$  automatically — no negative bikers, ever.
- One unit of  $x_j$ , rest fixed: the expected count is *multiplied* by  $e^{\beta_j}$  — the **rate ratio** ( $> 1$  more bikers,  $< 1$  fewer).
- The variance follows along:  $\text{Var}(Y_i) = \mu_i$  — busy hours are automatically noisier.

## Takeaway

Coefficients act on the **log of the mean**; effects on the count scale are **multiplicative**, never additive.

## Fitting it in statsmodels

```
import numpy as np, pandas as pd
import statsmodels.api as sm, statsmodels.formula.api as smf

bike = pd.read_csv('Bikeshare.csv', index_col=0) # 8 645 hourly rows
bike['hr'] = bike['hr'].astype(int) # hour 0..23
form = 'bikers ~C(hr) + temp + workingday + C(weathersit)'
pois = smf.glm(form, data=bike,
               family=sm.families.Poisson()).fit() # log link = canonical
print(pois.deviance, pois.df_resid) # 269 006 on 8 616 df
print(np.exp(pois.params['temp'])) # rate ratio 4.79
```

### Run this live in the lab notebook

Notebook `advanced_04_glms_splines_lab.ipynb`, §3 *Poisson regression on Bikeshare*: runs this fit, prints the full summary table and reproduces every rate ratio quoted on the next slide.

## Reading the fitted coefficients as rate ratios

| Term                       | $\hat{\beta}$          | SE    | $e^{\hat{\beta}}$ (rate ratio)  |
|----------------------------|------------------------|-------|---------------------------------|
| Intercept                  | 3.042                  | 0.009 | 20.9 bikers (baseline hour)     |
| temp                       | 1.567                  | 0.005 | 4.79 per full unit of temp      |
| workingday                 | 0.002                  | 0.002 | 1.002                           |
| weathersit=cloudy/misty    | -0.051                 | 0.002 | 0.951                           |
| weathersit=light rain/snow | -0.505                 | 0.004 | 0.603                           |
| weathersit=heavy rain/snow | -1.349                 | 0.167 | 0.260 (!) — from $n = 1$ hour   |
| hr=8 / hr=17 / hr=4        | 1.826 / 1.970 / -2.045 |       | 6.21 / 7.17 / 0.129 vs midnight |

## Three concrete readings

- **Light rain** cuts the expected count to  $e^{-0.505} = 0.603$  of the clear-sky rate — **39.7% fewer riders**, all else equal.
- $+0.1$  temp ( $\approx 4^\circ\text{C}$ ) multiplies the rate by  $e^{0.1 \times 1.567} = 1.17$  — **+17% riders**.
- Rush hour vs night:  $e^{1.970 - (-2.045)} = e^{4.015} = 55.4$  — 17:00 carries 55 $\times$  the 04:00 rate.

## Takeaway

Report  $e^{\hat{\beta}}$ , not  $\hat{\beta}$  — and distrust coefficients built on one observation (heavy rain)

# Offsets: modelling rates when exposure varies

## Rule (offset)

If  $Y_i$  is a count accumulated over exposure  $t_i$  (hours open, policy-years, population), model the *rate*  $\mu_i/t_i$ :

$$\log \frac{\mu_i}{t_i} = x_i^\top \beta \iff \log \mu_i = \underbrace{\log t_i}_{\text{offset: coefficient fixed at 1}} + x_i^\top \beta.$$

## Why raw counts mislead

Station A: 30 rentals in 2 hours  $\Rightarrow$  15.0/hour. Station B: 60 rentals in 6 hours  $\Rightarrow$  10.0/hour. B has *twice* A's count but only  $\frac{2}{3}$  of its rate; the offset makes the GLM compare 15.0 vs 10.0 — the comparison you actually want.

## Takeaway

An offset is a covariate with  $\beta$  **frozen at 1**: `smf.glm(..., offset=np.log(t))`. (Bikeshare needs none — every row is exactly one hour.)

## In industry — Insurance and epidemiology: the offset is the law of the land

Claim counts per policy-year and disease counts per 100 000 population are *always* fitted with exposure offsets — otherwise big portfolios and big cities dominate for size alone.

## Exercise A4.3 — Rate ratios on Bikeshare [Python]

### Exercise

Using the fitted Poisson GLM (`bikers ~ C(hr) + temp + workingday + C(weathersit)`):

- 1 Compute the rate ratio for `light` `rain/snow` versus `clear`, and phrase it as a percentage change in expected riders.
- 2 The `temp` coefficient is  $\hat{\beta} = 1.567$ . By what factor does the expected count change when `temp` rises by 0.2 (about 8°C)? Write the one line of Python that computes it from `pois.params`.
- 3 Why would it be wrong to say “a 0.2 rise in temp *adds*  $0.2 \times 1.567 = 0.31$  bikers”?

### Run this live in the lab notebook

The worked solution sits in `advanced_04_glms_splines_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Exercise A4.3 — Rate ratios on Bikeshare*.

## Solution — Exercise A4.3

### Solution

**Method.** All effects live on the log-mean scale; exponentiate to get multiplicative effects on the count scale.

**Part 1.**  $\hat{\beta}_{\text{light rain}} = -0.505$ , so the rate ratio is  $e^{-0.505} = 0.603$ : light rain/snow is associated with  $1 - 0.603 = 39.7\%$  fewer expected riders than clear weather, holding hour, temperature and day type fixed.

**Part 2.** Factor =  $e^{0.2 \times 1.567} = e^{0.313} = 1.368$  — about +36.8%.

```
float(np.exp(0.2 * pois.params['temp'])) # 1.368
```

**Part 3.** 0.31 is a change in  $\log \mu$ , not in  $\mu$ . The additive reading confuses the two scales: the actual change in expected riders depends on the starting level — +36.8% of 10 riders is  $\approx 3.7$  riders, of 400 riders  $\approx 147$ . Multiplicative effects have no single additive equivalent.

### Common mistake

Treating GLM coefficients like OLS slopes. On the log link,  $\hat{\beta}_j$  is a **log rate ratio**; only  $e^{\hat{\beta}_j}$  has a direct count-scale meaning.

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# Deviance: the GLM's residual sum of squares

## Rule (deviance)

With  $\hat{\ell}_{\text{sat}}$  the log-likelihood of the *saturated* model (one parameter per observation, each  $y_i$  fitted exactly):

$$D = 2 (\hat{\ell}_{\text{sat}} - \hat{\ell}) \quad (\phi = 1; \text{Poisson: } D = 2 \sum_i [y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i)]).$$

## How to read this

- Smaller  $D$  = closer to a perfect fit; for the Gaussian,  $D$  is the RSS — deviance generalises Chapter 3.
- The **null deviance**  $D_0$  (intercept-only model) plays the role of the total sum of squares.
- Bikeshare:  $D_0 = 1\,052\,921$  on 8 644 df  $\rightarrow D = 269\,006$  on 8 616 df — the 28 predictors remove  $\sim 74\%$  of the null deviance.

## Takeaway

Deviance is to GLMs what RSS is to OLS: the currency in which fits are compared.

# Likelihood-ratio tests: comparing nested GLMs

## Rule (LRT)

For nested models (reduced  $\subset$  full, differing by  $q$  parameters), under  $H_0$  “the extra parameters are zero”:

$$\text{LR} = D_{\text{reduced}} - D_{\text{full}} = 2(\hat{\ell}_{\text{full}} - \hat{\ell}_{\text{reduced}}) \stackrel{H_0}{\sim} \chi_q^2 \quad (n \rightarrow \infty).$$

### Does weather matter once hour + temp + day type are in?

Drop `weathersit` ( $q = 3$  dummies):  $D_{\text{reduced}} = 287\,263$ ,  $D_{\text{full}} = 269\,006$ , so  $\text{LR} = 18\,256.8$  on 3 df against a 5% critical value of  $\chi_{3,0.95}^2 = 7.81$ ;  $p$  is numerically 0. Weather stays in.

### Takeaway

The GLM analogue of Chapter 3’s  $F$ -test for a group of coefficients — same logic,  $\chi^2$  instead of  $F$ .

### Common mistake

The  $\chi^2$  result needs *nested* models fitted to the *same rows* by *maximum likelihood* — it does not apply to quasi-likelihood fits.

# AIC: model selection beyond nesting (bridge to Ch. 6)

## Rule (Akaike information criterion)

$AIC = -2\hat{\ell} + 2p$  (smaller is better;  $p$  = number of parameters).

### On Bikeshare

Full Poisson model:  $\hat{\ell} = -161\,022$ ,  $p = 29 \Rightarrow AIC = 322\,102$ . Without `weathersit`:  $AIC = 340\,352$ . The full model wins by 18 251 AIC points — consistent with the LRT.

### How to read this

- Chapter 6's AIC for regression was the Gaussian special case of this one.
- AIC compares **non-nested** models too (Poisson-with-splines vs Poisson-with-dummies) — as long as both are true likelihoods *for the same response data*.

### Takeaway

Deviance for nested tests, AIC for menus of candidates — both are likelihood arithmetic, both carry over to GAMs.

## Exercise A4.4 — Deviance, LRT and AIC by hand [Math]

### Exercise

For the Bikeshare Poisson GLM you are given: full model  $D = 269\,005.9$  on 8 616 df,  $\hat{\ell} = -161\,021.8$ ,  $p = 29$ ; reduced model (no `weathersit`)  $D = 287\,262.7$ ,  $p = 26$ .

- 1 Compute the likelihood-ratio statistic for `weathersit` and its degrees of freedom.
- 2 Compare with  $\chi^2_{3,0.95} = 7.81$  and state the conclusion.
- 3 Compute the full model's AIC from  $\hat{\ell}$  and  $p$ .
- 4 The reduced model's AIC is 340 352.4. Recover its log-likelihood.

## Solution — Exercise A4.4

## Solution

**Method.** Deviance differences are likelihood-ratio statistics; AIC is  $-2\hat{\ell} + 2p$  — everything is plug-in arithmetic.

**Part 1.**  $LR = D_{\text{red}} - D_{\text{full}} = 287\,262.7 - 269\,005.9 = 18\,256.8$ , with  $q = 29 - 26 = 3$  df (three weather dummies).

**Part 2.**  $18\,256.8 \gg 7.81$ : reject  $H_0$  decisively — weather adds explanatory power far beyond chance, given hour, temperature and day type.

**Part 3.**  $AIC = -2(-161\,021.8) + 2 \times 29 = 322\,043.7 + 58.0 = 322\,101.7$  ✓ (matches statsmodels).

**Part 4.**  $\hat{\ell}_{\text{red}} = -(AIC - 2p)/2 = -(340\,352.4 - 52)/2 = -170\,150.2$ . Check:  $2(\hat{\ell}_{\text{full}} - \hat{\ell}_{\text{red}}) = 2(-161\,021.8 + 170\,150.2) = 18\,256.8$  ✓ — the same LR as Part 1.

## Common mistake

Comparing AICs of models fitted to **different responses** (e.g.  $y$  vs  $\log y$ ) or different subsets of rows. AIC differences are only meaningful on identical data.

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- 2 Exponential Family
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# The Poisson straitjacket: variance = mean, no exceptions

The Poisson family has **no free variance parameter**: choosing it *asserts*  $\text{Var}(Y_i) = \mu_i$ .

## Bikeshare violates this on every shelf

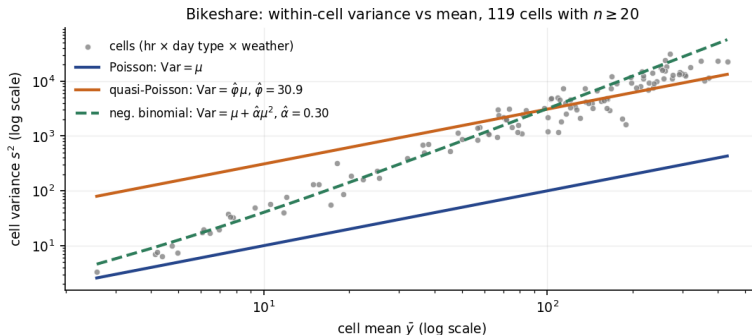
Group the hours into cells of (hour  $\times$  day type  $\times$  weather), keep the 119 cells with  $n \geq 20$ :

- 5 p.m., working day, clear: mean 431.5, variance 23 085 — ratio 53.5.
- 4 a.m., working day, clear: mean 4.97, variance 7.4 — ratio 1.5.
- **All 119 cells** have variance above their mean; the median ratio is 27.7.

## Takeaway

Counts in the wild are almost always **overdispersed**: unmodelled heterogeneity (events, marketing, school holidays...) inflates the variance beyond Poisson. The mean model can still be fine — the *uncertainty* is not.

# Seeing the violation: variance vs mean across cells



## Takeaway

The Poisson line  $\text{Var} = \mu$  runs far below every cell. The quasi-Poisson line  $\hat{\phi}\mu$  with  $\hat{\phi} = 30.9$  matches mid-range cells; the negative-binomial curve  $\mu + 0.30\mu^2$  bends up to catch the busy, high-variance cells. The data choose the curved law.

## Fix 1 — quasi-Poisson: rescale the uncertainty

### Rule (quasi-likelihood with dispersion)

Keep the mean model and  $V(\mu) = \mu$ , but allow  $\text{Var}(Y_i) = \varphi \mu_i$ , estimating

$$\hat{\varphi} = \frac{X^2}{n-p} = \frac{1}{n-p} \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}, \quad \widehat{\text{SE}}_{\text{quasi}}(\hat{\beta}_j) = \sqrt{\hat{\varphi}} \widehat{\text{SE}}_{\text{Pois}}(\hat{\beta}_j).$$

### Bikeshare

$X^2 = 266\,229$  on  $n - p = 8\,616$  df  $\Rightarrow \hat{\varphi} = 30.90$ : every standard error grows by  $\sqrt{30.90} = 5.56$ . **Point estimates do not move.** Consequences: temp z:  $330 \rightarrow 59.4$  (still overwhelming); workingday z:  $1.19 \rightarrow 0.21$  — its apparent effect evaporates.

### Takeaway

Overdispersion rarely changes *what* the effects are — it changes *how sure you are*. Borderline stars are the first casualties.

## Fix 2 — negative binomial: model the extra variance

### Rule (NB2 negative binomial)

Let  $Y_i | G_i \sim \text{Poisson}(\mu_i G_i)$  with a gamma-distributed multiplicative effect  $G_i$  ( $E[G_i] = 1$ ,  $\text{Var}(G_i) = \alpha$ ). Then  $E[Y_i] = \mu_i$ ,  $\text{Var}(Y_i) = \mu_i + \alpha \mu_i^2$ .

#### Bikeshare

ML estimate  $\hat{\alpha} = 0.305$ . Log-likelihood improves from  $-161\,022$  (Poisson) to  $-45\,266$  (NB); AIC drops from  $322\,102$  to  $90\,590$  — not a close call. At  $\mu = 400$  the implied variance is  $400 + 0.305 \times 400^2 \approx 49\,200$ , versus Poisson's 400.

#### Quasi vs NB in one line each

*Quasi-Poisson*: same estimates, honest SEs, **no likelihood** (no AIC, no LRT). *Negative binomial*: a genuine likelihood — estimates shift (it reweights observations), AIC/LRT available, variance grows *quadratically*.

#### Takeaway

Only need honest standard errors? Quasi suffices. Need likelihood machinery or predictive distributions? Fit the negative binomial.

## Poisson vs quasi-Poisson vs negative binomial, side by side

| Term            | Poisson       |       | quasi-Poisson        | Negative binomial |        |
|-----------------|---------------|-------|----------------------|-------------------|--------|
|                 | $\hat{\beta}$ | SE    | SE ( $\times 5.56$ ) | $\hat{\beta}$     | SE     |
| temp            | 1.567         | 0.005 | 0.026                | 1.894             | 0.032  |
| workingday      | 0.002         | 0.002 | 0.011                | -0.195            | 0.013  |
| light rain/snow | -0.505        | 0.004 | 0.022                | -0.538            | 0.022  |
| AIC             | 322 102       |       | — (no likelihood)    |                   | 90 590 |

## Which numbers changed — and why

Quasi keeps every  $\hat{\beta}$  and multiplies SEs by  $\sqrt{\hat{\phi}} = 5.56$ . NB *refits*: its likelihood downweights huge counts, so estimates move — *workingday* even flips sign, because its true effect is heterogeneous across hours (positive at commuter peaks, negative at night) and the two likelihoods average that heterogeneity with different weights.

## In industry — Retail demand: the dispersion audit

Demand-forecasting teams sanity-check  $\hat{\phi}$  per store cluster. A jump of  $\hat{\phi}$  from 3 to 30 after a promotion week is a red flag that the mean model is missing a driver — not a licence for tighter intervals.

## Exercise A4.5 — Dispersion arithmetic [Math]

### Exercise

The Bikeshare Poisson fit has Pearson statistic  $X^2 = 266\,229$  with  $n = 8\,645$  observations and  $p = 29$  parameters. The `temp` coefficient is  $\hat{\beta} = 1.567$  with Poisson SE 0.0047.

- 1 Estimate the dispersion  $\hat{\phi}$  and the factor by which all standard errors must grow.
- 2 Compute the 95% confidence interval for  $\beta_{\text{temp}}$  under the Poisson fit and under quasi-Poisson (use  $\pm 1.96$  SE).
- 3 Convert both intervals to rate-ratio scale.
- 4 A colleague argues: “the quasi interval is wider, so the Poisson one is more precise — keep it.” Rebut in one sentence.

## Solution — Exercise A4.5

### Solution

**Method.**  $\hat{\varphi} = X^2/(n - p)$ ; quasi SEs scale by  $\sqrt{\hat{\varphi}}$ ; exponentiate CI endpoints for rate ratios.

**Part 1.**  $\hat{\varphi} = 266\,229/(8\,645 - 29) = 266\,229/8\,616 = 30.90$ ; SEs grow by  $\sqrt{30.90} = 5.56$ .

**Part 2.** Poisson:  $1.567 \pm 1.96 \times 0.0047 = [1.558, 1.576]$ . Quasi:  $SE = 5.56 \times 0.0047 = 0.0264$ , so  $1.567 \pm 1.96 \times 0.0264 = [1.515, 1.619]$ .

**Part 3.** Rate-ratio scale: Poisson  $[e^{1.558}, e^{1.576}] = [4.75, 4.83]$ ; quasi  $[e^{1.515}, e^{1.619}] = [4.55, 5.05]$ .

**Part 4.** The Poisson interval is not *more precise*, it is *wrong*: its claimed coverage assumes  $\text{Var} = \mu$ , which the data contradict by a factor of  $\sim 31$ , so the narrow interval covers the truth far less often than 95%.

### Common mistake

Reading interval width as a quality metric. Width is only meaningful *after* the variance assumption survives a check like  $\hat{\varphi} \approx 1$ .

## Extended Exercise A4.1 — The overdispersion pipeline [Integrative]

### Extended exercise (15 min) — from suspicion to model choice

Work through the full overdispersion protocol on Bikeshare ( $\text{bikers} \sim \text{C}(\text{hr}) + \text{temp} + \text{workingday} + \text{C}(\text{weathersit})$ ):

- 1 **Diagnose.** Compute  $\hat{\phi}$  from the Poisson fit's Pearson statistic. What values would have let you keep plain Poisson?
- 2 **Repair the SEs.** Which of the following change under quasi-Poisson, and how: point estimates, standard errors, fitted means, AIC?
- 3 **Re-model.** The negative binomial fit gives  $\hat{\alpha} = 0.305$  and AIC 90 590 vs Poisson's 322 102. Why is comparing these two AICs legitimate, while “quasi-Poisson AIC” does not exist?
- 4 **Decide.** *workingday* has  $\hat{\beta} = 0.002$  (Poisson),  $z_{\text{quasi}} = 0.21$ , but  $\hat{\beta} = -0.195$  (NB,  $z \approx -14.7$ ). What should an analyst conclude and report about the working-day effect?

### Run this live in the lab notebook

Notebook §5 *Overdispersion: quasi-Poisson and negative binomial* fits all three models and prints the comparison table from the previous slide.

## Solution — Extended Exercise A4.1 (1/2)

### Solution — diagnosis and repair

**Method.** One dispersion statistic drives the whole protocol; then separate what quasi-likelihood rescales from what a new likelihood refits.

**Part 1.**  $\hat{\phi} = 266\,229/8\,616 = 30.90$ . Values near 1 (rule of thumb  $\lesssim 1.5$ , given the huge  $n$  here even tighter) would justify plain Poisson; 30.9 is a categorical rejection — the variance is  $\sim 31\times$  the Poisson claim on average.

**Part 2.** Under quasi-Poisson: point estimates **unchanged** (same estimating equations); fitted means **unchanged**; standard errors **multiplied by**  $\sqrt{30.90} = 5.56$ ; AIC **undefined** — quasi-likelihood is not a likelihood, so there is no  $\hat{\ell}$  to plug into  $-2\hat{\ell} + 2p$ .

## Solution — Extended Exercise A4.1 (2/2)

### Solution — re-model and decide

**Part 3.** Poisson and negative binomial are both *genuine likelihoods for the same response vector*, so their AICs are comparable: the NB's  $90\,590 \ll 322\,102$  says the quadratic variance law fits these counts far better. Quasi-Poisson specifies only mean and variance, never a density, hence no likelihood and no AIC.

**Part 4.** The honest conclusion: *averaged over all hours*, working days do not add a stable multiplicative effect — quasi says the average effect is indistinguishable from zero, and the NB's  $-0.195$  reflects its different weighting of hours, not a newly discovered negative effect. The effect is **heterogeneous** (strongly positive at 8 a.m. and 5–6 p.m., negative at night and midday); report it via an *interaction* ( $\text{workingday} \times \text{hour}$ ) or a GAM with separate hour curves, not a single number.

### Common mistake

Reading the sign flip as “the NB found the true effect”. When two reasonable likelihoods disagree this much, the model is mis-specified — the fix is a richer mean model, not a coin flip between packages.

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## Chapter 7 recap: regression splines and the knot dilemma

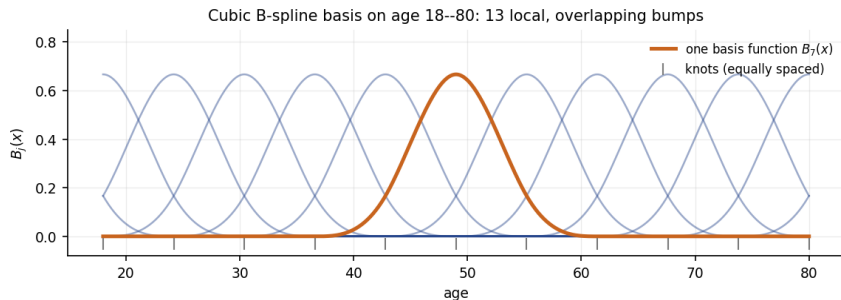
### What you already know

- A **regression spline** of degree 3 is a piecewise cubic, forced to be continuous with continuous first and second derivatives at the **knots**.
- Fit by OLS on a spline **basis**: flexibility is controlled by the *number and placement of knots* (equivalently, the degrees of freedom).
- Chapter 7's recipe: place knots at quantiles, pick df by cross-validation.

### The dilemma this section resolves

Knot choice is **discrete and awkward**: 5 vs 6 knots is a lumpy, all-or-nothing decision, and knots in the wrong place cannot be undone by fitting. The penalized view replaces “how many knots?” by a single **continuous** dial  $\lambda$ : use *plenty* of knots, then penalise wiggleness.

# The B-spline basis: local, overlapping bumps



## Takeaway

Each cubic B-spline  $B_j(x)$  is positive on only 4 adjacent knot intervals and zero elsewhere; at any  $x$  they sum to 1. The fit  $f(x) = \sum_j \beta_j B_j(x)$  is a **local** blend — moving one  $\beta_j$  reshapes the curve only near its bump, which is why the basis is numerically stable where raw polynomials ( $x^7, x^8, \dots$ ) explode.

# Penalized splines: many knots, one dial

## Rule (penalized fitting)

Use a rich B-spline basis (20–40 knots) and minimise

$$\underbrace{\text{Deviance}(\beta)}_{\text{fit the data}} + \lambda \underbrace{\int f''(t)^2 dt}_{\text{punish wiggleness}} \quad \text{with} \quad \int f''(t)^2 dt = \beta^\top S \beta.$$

Gaussian case:  $\hat{\beta}_\lambda = (B^\top B + \lambda S)^{-1} B^\top y$  — ridge regression (Ch. 6) with a structured penalty.

### How to read this

$\lambda = 0$ : unpenalized — as wiggly as the basis allows.  $\lambda \rightarrow \infty$ :  $f'' \equiv 0$  forced — a **straight line** (zero curvature is invisible to the penalty). In between: the bias–variance dial of Chapter 2, turned *continuously*.

### Takeaway

Knot placement stops mattering once knots are plentiful; **all** structural choice is concentrated in the single number  $\lambda$ .

# Effective degrees of freedom: how much did you really fit?

## Rule (edf)

The penalized fit is linear in  $y$ :  $\hat{y} = H(\lambda)y$  with smoother matrix  $H(\lambda) = B(B^\top B + \lambda S)^{-1}B^\top$ . Define

$$\text{edf}(\lambda) = \text{tr}\{H(\lambda)\} = \sum_k \frac{1}{1 + \lambda d_k}, \quad d_k \geq 0 \text{ the penalty's eigenvalues in the basis.}$$

## Toy computation

Four basis directions with  $d = (0, 0, 1, 4)$  and  $\lambda = 1$ :  $\text{edf} = 1 + 1 + \frac{1}{2} + \frac{1}{5} = 2.7$ . The two  $d = 0$  directions (constant, linear — zero curvature) are never shrunk; the wiggly ones are partially switched off.

## Takeaway

edf interpolates smoothly between 2 (a line) and  $K$  (the full basis) — “this smooth is worth 6.3 parameters” is a meaningful, comparable statement.

# Choosing $\lambda$ : cross-validation or GCV (bridge to Ch. 5)

## Rule (generalized cross-validation)

$$\text{GCV}(\lambda) = \frac{n \text{RSS}(\lambda)}{(n - \text{edf}(\lambda))^2} \quad \text{— minimise over } \lambda.$$

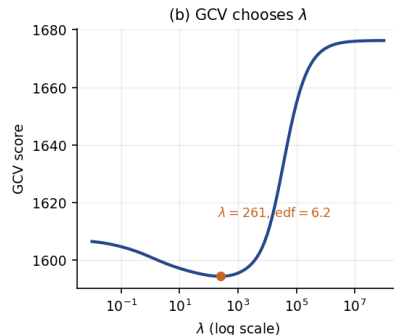
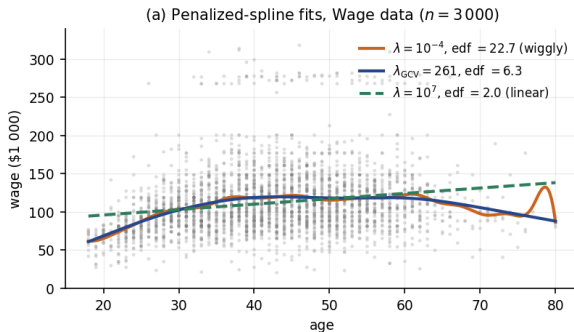
### How to read this

- GCV is a shortcut to leave-one-out CV: no  $n$  refits, just the single fit's edf — the same idea as Chapter 5's LOOCV formula for linear smoothers.
- $k$ -fold CV on a  $\lambda$ -grid works too — the honest default when the model is not a linear smoother (e.g. GLM deviance losses).
- Watch **edf**, not  $\lambda$ :  $\lambda$ 's scale depends on the basis, edf does not.

### Takeaway

$\lambda$  is a tuning parameter like any in Chapters 5–6: chosen by (generalized) cross-validation, never by eye.

# One picture: Wage across three orders of magnitude of $\lambda$



## Takeaway

23 cubic B-splines on age ( $n = 3000$ ). At  $\lambda = 10^{-4}$  the fit burns 22.7 edf and chases noise around age 78; at  $\lambda = 10^7$  it collapses to a line (2.0 edf) and misses the plateau. GCV picks  $\lambda = 261$ , edf = 6.3 — close to the df Chapter 7 found by CV for this very curve.

## Exercise A4.6 — Reading the $\lambda$ dial [Math]

### Exercise

A penalized cubic spline uses  $K = 23$  B-spline basis functions with a second-derivative penalty  $S$  whose null space contains exactly the constant and linear functions.

- 1 State edf in the limits  $\lambda \rightarrow 0$  and  $\lambda \rightarrow \infty$ , with one sentence of justification each.
- 2 In a small example the penalty eigenvalues are  $d = (0, 0, 1, 4)$ . Compute edf for  $\lambda = 1$  and for  $\lambda = 4$ .
- 3 Your colleague reports “ $\lambda = 261$ ” as the headline of their smooth. Why is “ $\text{edf} = 6.3$ ” the better headline?

## Solution — Exercise A4.6

### Solution

**Method.** Use  $\text{edf}(\lambda) = \sum_k 1/(1 + \lambda d_k)$  and think about which directions the penalty can and cannot touch.

**Part 1.**  $\lambda \rightarrow 0$ : no shrinkage, every basis direction survives,  $\text{edf} \rightarrow K = 23$  (an unpenalized regression spline).  $\lambda \rightarrow \infty$ : every direction with  $d_k > 0$  is annihilated; only the null space of  $S$  — constant + linear — survives, so  $\text{edf} \rightarrow 2$  (a straight line).

**Part 2.**  $\lambda = 1$ :  $1 + 1 + \frac{1}{1+1} + \frac{1}{1+4} = 2 + 0.5 + 0.2 = 2.70$ .

$\lambda = 4$ :  $1 + 1 + \frac{1}{5} + \frac{1}{17} = 2 + 0.200 + 0.059 = 2.26$  — quadrupling  $\lambda$  removed less than half an edf: the dial is logarithmic.

**Part 3.**  $\lambda$ 's numerical value depends on the basis size, the knot spacing, even the units of  $x$  — 261 is meaningless across studies.  $\text{edf}$  is invariant to all of that and reads directly as model complexity: “a curve worth 6.3 parameters”.

### Common mistake

Comparing  $\lambda$  values across different bases or datasets. Only **edf** travels.

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# The meeting point: GAM = GLM + smooth terms

## Rule (penalized-spline GAM for counts)

$$Y_i \sim \text{Poisson}(\mu_i), \quad \log \mu_i = \beta_0 + f_1(\text{hr}_i) + f_2(\text{temp}_i) + \beta_w \text{workingday}_i + \sum_k \gamma_k \text{weather}_i$$

each  $f_j$  a penalized B-spline with its own  $\lambda_j$ ; fit by **penalized maximum likelihood** (penalized IRLS).

## How to read this

- Everything from this module composes: exponential family (random part), log link, deviance-based fitting, roughness penalties per smooth.
- Chapter 7 introduced GAMs with fixed-df smoothers; the upgrade here is that each curve's complexity is *estimated* via  $\lambda_j$ , not preset.
- Additivity is retained: each  $f_j$  is a one-dimensional curve you can *plot and defend* — the interpretability contract of Ch. 7 survives.

# Fitting the GAM with statsmodels

```
from statsmodels.gam.api import GLMGam, BSplines

xs = bike[['hr', 'temp']].astype(float)
bs = BSplines(xs, df=[12, 6], degree=[3, 3]) # rich bases for hr, temp
Xpar = ... # intercept, workingday, 3 weather dummies (parametric part)

gam = GLMGam(bike.bikers, exog=Xpar, smoother=bs,
             family=sm.families.Poisson(), alpha=[0.319, 0.00423])
res = gam.fit()
# the two penalty weights above were chosen by the built-in criterion:
# gam.select_penweight()[0] -> [0.319, 0.00423]
print(res.deviance) # 259 650
```

## Run this live in the lab notebook

Notebook §7 A penalized-spline GAM for Bikeshare builds Xpar, runs select\_penweight, and draws the partial-effect plots on the next slide with confidence bands.

# What the machine chose

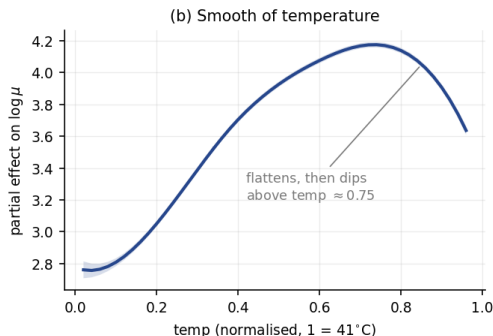
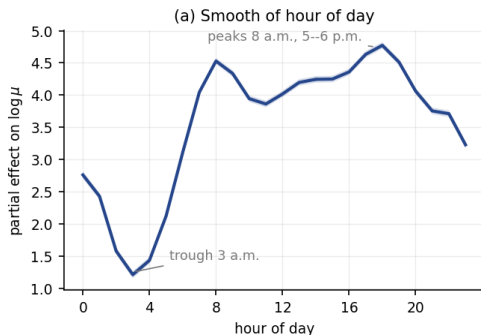
## Selected penalties and complexity on Bikeshare

- `select_penweight` (GCV-type criterion) returns  $\hat{\alpha} = (0.319, 0.00423)$  for (hr, temp).
- Resulting complexity: hour smooth  $\approx 11$  edf, temperature smooth  $\approx 5$  edf — the light penalties keep almost all of the 12 and 6 basis df: with 8 645 observations the data can afford it.
- Deviance 259 650 vs 269 006 for the dummy-hr GLM with linear temp; AIC 312 730 vs 322 102 — the smooth-temp model wins by  $\sim 9\,400$  AIC points with *fewer* effective parameters.

## Takeaway

The gain comes almost entirely from letting temp **bend**: the linear-temp assumption of the earlier GLM was the binding constraint, not the hour dummies.

# The fitted smooths: hour of day and temperature



## Takeaway

**Hour:** trough at 3 a.m., twin commuter peaks at 8 a.m. and 5–6 p.m.; peak-to-trough ratio  $e^{4.77-1.21} = 35\times$ . **Temp:** riders triple ( $\times 3.1$ ) from temp 0.2 to 0.7, then *fall* beyond  $\approx 0.75$  ( $\sim 31^\circ\text{C}$ ) — a hot-weather dip a linear term is structurally unable to show.

# Interpreting the hour curve — and one honest caveat

## Reading a partial effect

Each curve shows the contribution to  $\log \mu$  *holding the other terms fixed*. Differences along one curve exponentiate to rate ratios:  $f(18) - f(3) = 4.77 - 1.21 = 3.56 \Rightarrow e^{3.56} \approx 35\times$  more riders at 6 p.m. than at 3 a.m., same weather, same temperature, same day type.

## Smooths smooth — also where the truth is spiky

The dummy model rates 17:00 at  $55.4\times$  the 04:00 level; the smooth version says  $24.6\times$ . The penalty shrinks the razor-sharp night dip and commuter spikes toward their neighbours. For genuinely *discontinuous* patterns (timetables, opening hours), factors or more basis df beat a heavily penalized smooth — check edf and residuals by hour before trusting the curve.

## In industry — Energy: GAMs run the grid

European system operators forecast national electricity load with exactly this structure — smooth(hour), smooth(temperature), calendar dummies — because engineers must *see and sign off* each curve. The hot-weather bend (air conditioning) is the industry's textbook example of why temp must be a smooth.

# Pitfalls when GLMs meet splines

## Don't

- ❶ **Extrapolate a spline.** Outside the knot range the fit continues as a low-order polynomial with *no data behind it*; on the log scale a mild linear overshoot exponentiates into absurd counts.
- ❷ **Read smooths causally.**  $f(\text{temp})$  is an association given the other terms; hot hours also differ in season, daylight and tourism. Nothing here is a do-operator.
- ❸ **Ignore overdispersion because the mean is now flexible.** The GAM's deviance is still Poisson: with  $\hat{\varphi} \approx 31$  on this data, its naive confidence bands are  $\sim 5.6\times$  too narrow too. Rescale or go NB.

## Takeaway

Flexibility fixes the *mean*; it does not buy causality and it does not fix the *variance*. Keep the three audits separate.

## Extended Exercise A4.2 — Build and defend a count GAM [Python]

### Extended exercise (15 min) — the full modelling arc

Using `GLMGam` with `BSplines(df=[12, 6], degree=[3, 3])` on `hr` and `temp`, plus parametric `workingday` and weather dummies:

- 1 Write down the model equation for  $\log \mu_i$  and say which parts are penalized.
- 2 The selected penalties give  $\text{edf} \approx 11$  (`hr`) and  $\approx 5$  (`temp`). What would  $\text{edf} \approx 2$  for `temp` have told you?
- 3 From the fitted hour smooth,  $f(18) = 4.77$  and  $f(3) = 1.21$ . Compute the 6 p.m. vs 3 a.m. rate ratio and interpret it in one sentence.
- 4 The GAM's weather coefficient is  $-0.569$  (light rain) vs the GLM's  $-0.505$ . Give one reason the estimates differ.
- 5 Name the single most important robustness check before this model ships to the fleet-rebalancing team.

### Run this live in the lab notebook

Notebook §7 closes with this exercise: the cell headed *Extended Exercise A4.2* reproduces every number.

## Solution — Extended Exercise A4.2 (1/2)

### Solution — model / edf / rate ratio

**Method.** Assemble the GAM equation, then read complexity from edf and effects from differences on the log scale.

**Part 1.**  $\log \mu_i = \beta_0 + f_1(\text{hr}_i) + f_2(\text{temp}_i) + \beta_w \text{workingday}_i + \gamma_1 \text{cloudy}_i + \gamma_2 \text{light}_i + \gamma_3 \text{heavy}_i$ . Only  $f_1$  and  $f_2$  are penalized (each with its own  $\alpha$ ); intercept, dummies and  $\beta_w$  are ordinary ML parameters.

**Part 2.**  $\text{edf} \approx 2$  for `temp` would mean the penalty had shrunk the smooth to (near-)linearity — i.e. the data see no curvature and the Ch. 4 linear-temp GLM would have been adequate. The actual  $\text{edf} \approx 5$  says the bend (including the hot-weather dip) is real structure worth  $\sim 5$  parameters.

**Part 3.**  $e^{4.77-1.21} = e^{3.56} \approx 35$ : holding temperature, weather and day type fixed, the expected count at 6 p.m. is about 35 times the 3 a.m. count.

## Solution — Extended Exercise A4.2 (2/2)

### Solution — coefficient shift and the shipping check

**Part 4.** The two models adjust for temperature differently: the GLM forces a linear temp effect, the GAM lets it bend. Rainy hours are also cool hours, so when the temperature adjustment changes shape, the weather dummies absorb a different share of the rain–cold overlap ( $-0.569$  vs  $-0.505$ ). Coefficients of correlated regressors always move when a co-regressor's functional form changes.

**Part 5.** An **overdispersion audit** of the GAM itself: compute the Pearson dispersion of the GAM fit (it remains  $\gg 1$  on this data) and widen the uncertainty accordingly (quasi rescaling or an NB-GAM). Shipping the  $35\times$  headline with Poisson bands would overstate certainty roughly 5–6-fold — the fleet team would over-trust hour-level differences that are partly noise.

### Common mistake

Declaring victory after the mean model improves. The variance audit (§Overdispersion) applies to GAMs verbatim — flexible means do not repair broken variances.

# Outline

- 1 Counts Break the LM
- 2 Exponential Family
- 3 GLM Anatomy
- 4 Inference
- 5 Overdispersion
- 6 Penalized Splines
- 7 GAMs for Counts
- 8 Summary**

## Module A4 in one slide

### What we accomplished

- Diagnosed why OLS fails on counts: negative means, constant variance — 10% negative predictions on Bikeshare.
- Unified Chs. 3–4 in the **exponential family**: mean  $b'(\theta)$ , variance  $b''(\theta)a(\phi)$ .
- Assembled the **GLM**: family + linear predictor + link; Poisson coefficients as **rate ratios**; offsets for exposure.
- Ran GLM **inference**: deviance, LRT (18 257 on 3 df for weather), AIC.
- Confronted **overdispersion** ( $\hat{\phi} = 30.9$ ): quasi-Poisson rescales SEs  $\times 5.56$ ; negative binomial ( $\hat{\alpha} = 0.305$ ) refits.
- Rebuilt splines as **penalized** fits: one dial  $\lambda$ , complexity read as edf, tuned by GCV/CV.
- Met in the middle: a **penalized-spline Poisson GAM** with interpretable hour and temperature curves.

# Key formulas at a glance

**Exponential family**  $f(y; \theta, \phi) = \exp\{(y\theta - b(\theta))/a(\phi) + c(y, \phi)\}$

**Mean and variance**  $E[Y] = b'(\theta)$ ,  $\text{Var}(Y) = b''(\theta) a(\phi) = a(\phi) V(\mu)$

**GLM link**  $g(\mu_i) = \eta_i = x_i^\top \beta$ ; canonical link  $g = (b')^{-1}$

**Poisson rate ratio** one unit of  $x_j$  multiplies  $\mu$  by  $e^{\beta_j}$ ; offset:  $\log \mu_i = \log t_i + x_i^\top \beta$

**Deviance / LRT**  $D = 2(\hat{\ell}_{\text{sat}} - \hat{\ell})$ ;  $D_{\text{red}} - D_{\text{full}} \sim \chi_q^2$  under  $H_0$

**AIC**  $-2\hat{\ell} + 2p$

**Dispersion**  $\hat{\phi} = X^2/(n-p)$ ; quasi SE =  $\sqrt{\hat{\phi}} \times \text{Poisson SE}$

**Negative binomial**  $\text{Var}(Y) = \mu + \alpha\mu^2$

**Penalized spline** minimise Deviance +  $\lambda \int f''(t)^2 dt = \text{Dev} + \lambda \beta^\top S \beta$

**Effective df**  $\text{edf} = \text{tr}\{B(B^\top B + \lambda S)^{-1} B^\top\} = \sum_k (1 + \lambda d_k)^{-1}$

**GCV**  $n\text{RSS}/(n - \text{edf})^2$

**Count GAM**  $\log \mu_i = \beta_0 + \sum_j f_j(x_{ij}) + (\text{parametric terms})$

# Vocabulary checklist

| Term                          | Meaning                                                              |
|-------------------------------|----------------------------------------------------------------------|
| Canonical parameter $\theta$  | natural scale on which $y$ enters the density linearly               |
| Cumulant function $b(\theta)$ | generates mean ( $b'$ ) and variance ( $b''$ )                       |
| Variance function $V(\mu)$    | mandated shape of variance as a function of the mean                 |
| Link function $g$             | maps the mean's range to $\mathbb{R}$ ; canonical if $g = (b')^{-1}$ |
| Linear predictor $\eta$       | $x^\top \beta$ , the part shared with Chs. 3–4                       |
| Rate ratio                    | $e^{\beta_j}$ : multiplicative effect on the expected count          |
| Offset                        | covariate with coefficient fixed at 1, encodes exposure              |
| Saturated model / deviance    | perfect-fit benchmark; $2 \times$ log-likelihood gap to it           |
| Overdispersion                | $\text{Var} > \text{family's claim}$ ; here $\hat{\phi} = 30.9$      |
| Quasi-likelihood              | mean–variance assumptions only; no density, no AIC                   |
| Negative binomial             | gamma-mixed Poisson; $\text{Var} = \mu + \alpha\mu^2$                |
| B-spline basis                | local overlapping polynomial bumps; numerically stable               |
| Roughness penalty             | $\lambda \int f''^2$ : price tag on curvature                        |
| Effective df (edf)            | $\text{tr } H(\lambda)$ : parameters actually spent                  |
| GCV                           | leave-one-out shortcut for linear smoothers                          |
| GAM / partial effect          | additive smooth terms inside a GLM; one curve per predictor          |

## Decision rules of thumb

- Response is a **count**  $\Rightarrow$  start with Poisson + log link; **binary**  $\Rightarrow$  Bernoulli + logit; **continuous symmetric**  $\Rightarrow$  Gaussian + identity.
- Counts accumulated over unequal windows  $\Rightarrow$  **offset**  $\log(\text{exposure})$ , always.
- After any Poisson fit, compute  $\hat{\phi} = X^2/(n - p)$ . If  $\hat{\phi} \gtrsim 1.5$ : quasi-Poisson for quick honest SEs; negative binomial when you need AIC, LRTs or predictive distributions.
- Effect shape unknown  $\Rightarrow$  penalized spline with generous basis; let GCV/CV set  $\lambda$ ; **report edf**, not  $\lambda$ .
- Smooth's edf  $\approx 2 \Rightarrow$  simplify to a linear term; edf near the basis size  $\Rightarrow$  consider a larger basis or a factor.
- Report  $e^{\hat{\beta}}$  (rate/odds ratios) with intervals from the *dispersion-corrected* fit.

### Takeaway

Family first, link second, dispersion audit third, smoothness by CV fourth — in that order, every time.

# Common pitfalls

## Don't

- 1 **Fit OLS to raw counts** (or to  $\log(y + 1)$  and call it equivalent) — wrong support, wrong variance, biased back-transformed means.
- 2 **Interpret GLM coefficients additively** — on a log link only  $e^{\hat{\beta}}$  speaks the count language.
- 3 **Trust Poisson standard errors without a dispersion check** — at  $\hat{\phi} = 30.9$  they are  $5.6\times$  too small; borderline “discoveries” evaporate.
- 4 **Use AIC/LRT on quasi-likelihood fits** — there is no likelihood there.
- 5 **Extrapolate splines beyond the knots** or compare raw  $\lambda$  values across bases — use edf and stay in range.
- 6 **Read GAM smooths as causal effects** — they are adjusted associations, nothing more.

# Self-check questions

## Can you answer these without the slides?

- 1 Derive the mean and variance of the Poisson from  $b(\theta) = e^\theta$ .
- 2 Why is the logit the *canonical* link for the Bernoulli, and what does canonicity buy?
- 3 Weather's LRT was 18 257 on 3 df. Which two models were compared, and what assumption makes the  $\chi^2$  reference valid?
- 4 Quasi-Poisson and negative binomial both "fix" overdispersion — name two practical situations that force the NB choice.
- 5 Why does  $\lambda \rightarrow \infty$  leave a straight line rather than a constant, for a second-derivative penalty?
- 6 Your GAM's temperature smooth has edf 5.0 and dips after temp 0.75. What would you check before presenting the dip as real?

## Takeaway

If question 1 takes more than two lines, revisit the worked Poisson derivation — it is the

## Appendix — what is in here, and why

Nothing in the appendix is needed to follow the chapter: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

### Contents of the appendix

- **Score identities** — the two-line proof of  $E[Y] = b'(\theta)$  and  $\text{Var}(Y) = b''(\theta)a(\phi)$ ; moved out because the main deck only needs the result.
- **The Gaussian as a family member** — completes the table of three; moved out because it mirrors the Poisson derivation done live.
- **IRLS in one slide** — how GLMs are actually fitted; moved out because no lecture decision depends on the algorithm's internals.
- **Deviance formulas per family** — reference table; moved out because it is look-up material, not narrative.
- **Drill exercise on offsets** — extra practice with full solution; moved out to keep the main flow at two extended exercises.

# Score identities: where mean and variance come from

For one observation,  $\ell(\theta) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)$ .

## Derivation

*First identity.* Under regularity,  $E[\partial\ell/\partial\theta] = 0$ :

$$E\left[\frac{Y - b'(\theta)}{a(\phi)}\right] = 0 \implies E[Y] = b'(\theta).$$

*Second identity.* The Bartlett identity  $E[\partial^2\ell/\partial\theta^2] + E[(\partial\ell/\partial\theta)^2] = 0$  gives

$$-\frac{b''(\theta)}{a(\phi)} + \frac{\text{Var}(Y)}{a(\phi)^2} = 0 \implies \text{Var}(Y) = b''(\theta) a(\phi). \quad \blacksquare$$

## Takeaway

Both facts used throughout the module are two lines of calculus on the score — no distribution-specific work needed.

# The Gaussian as a family member (completing the table)

## Match the normal density to the family form

$$f(y; \mu, \sigma^2) = \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right\}.$$

Matching:  $\theta = \mu$ ,  $b(\theta) = \theta^2/2$ ,  $a(\phi) = \sigma^2$ ,  $c(y, \phi) = -y^2/(2\sigma^2) - \frac{1}{2} \log(2\pi\sigma^2)$ .

Identities:  $E[Y] = b'(\theta) = \theta = \mu$  ✓;  $\text{Var}(Y) = b''(\theta)a(\phi) = 1 \cdot \sigma^2$  ✓.

## Takeaway

$b'' \equiv 1$ : the Gaussian is the *only* member whose variance ignores the mean — exactly the property that made OLS fail on counts, and exactly why Chapter 3 never needed a variance function.

# IRLS: how a GLM is actually fitted

## Algorithm (iteratively reweighted least squares)

Repeat until convergence, starting from any sensible  $\hat{\mu}^{(0)}$ :

- 1 Working response:  $z_i = \hat{\eta}_i + (y_i - \hat{\mu}_i) g'(\hat{\mu}_i)$ .
- 2 Working weights:  $w_i = [g'(\hat{\mu}_i)^2 V(\hat{\mu}_i)]^{-1}$ .
- 3 Update  $\beta$  by weighted least squares of  $z$  on  $X$  with weights  $w$ ; set  $\hat{\eta} = X\hat{\beta}$ ,  $\hat{\mu} = g^{-1}(\hat{\eta})$ .

## How to read this

- IRLS is Newton–Raphson with the expected information (Fisher scoring), rearranged as repeated WLS — which is why GLM software is built on the Chapter 3 machinery.
- Penalized GAM fitting only changes step 3: WLS becomes *penalized* WLS with  $\lambda S$  added — the ridge update from Chapter 6.

## Takeaway

A GLM is “OLS in a loop”; a GAM is “ridge in a loop”. Nothing beyond this course’s toolbox.

## Deviance formulas per family (reference)

| Family        | Unit deviance contribution $d_i$                                                                                    | Note                                         |
|---------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Gaussian      | $(y_i - \hat{\mu}_i)^2$                                                                                             | $D = \text{RSS}$ : Ch. 3 recovered           |
| Bernoulli     | $-2[y_i \log \hat{\mu}_i + (1 - y_i) \log(1 - \hat{\mu}_i)]$                                                        | = the cross-entropy loss                     |
| Poisson       | $2[y_i \log(y_i / \hat{\mu}_i) - (y_i - \hat{\mu}_i)]$                                                              | terms with $y_i = 0$ : $2\hat{\mu}_i$        |
| Neg. binomial | $2[y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i + \frac{1}{\alpha}) \log \frac{1 + \alpha y_i}{1 + \alpha \hat{\mu}_i}]$ | reduces to Poisson as $\alpha \rightarrow 0$ |

## How to read this

$D = \sum_i d_i$  in each case. Residual checks for GLMs use *deviance residuals*  $\text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$ , which are the closest GLM analogue of the OLS residuals from Chapter 3.

# Drill Exercise A4.A — Offsets in anger [Math]

## Exercise

A city compares cycling at two counting stations using a Poisson GLM with log link. Station 1 recorded  $y_1 = 180$  riders over  $t_1 = 12$  hours; Station 2 recorded  $y_2 = 96$  riders over  $t_2 = 4$  hours.

- 1 Compute each station's empirical rate per hour.
- 2 In the model  $\log \mu_i = \log t_i + \beta_0 + \beta_1 \mathbb{I}\{i = 2\}$ , express  $e^{\beta_1}$  in terms of the two rates and compute its ML estimate.
- 3 What would the same model *without* the offset (raw counts) have estimated for  $e^{\beta_1}$ , and why is that misleading?

# Solution — Drill Exercise A4.A

## Solution

**Method.** With a saturated two-group Poisson model, ML fits each group's rate exactly — so everything reduces to ratios of empirical rates.

**Part 1.** Station 1:  $180/12 = 15.0$  riders/hour. Station 2:  $96/4 = 24.0$  riders/hour.

**Part 2.** With the offset,  $\mu_i = t_i e^{\beta_0 + \beta_1 \mathbb{1}_{\{i=2\}}}$ , so  $e^{\beta_0}$  is Station 1's rate and  $e^{\beta_1}$  the **rate ratio**:

$$e^{\hat{\beta}_1} = \frac{y_2/t_2}{y_1/t_1} = \frac{24.0}{15.0} = 1.60$$

— Station 2 sees 60% *more* riders per hour.

**Part 3.** Without the offset the model compares raw counts:  $e^{\hat{\beta}_1} = 96/180 = 0.533$  — Station 2 would look 47% *quieter*, purely because it was observed for a third of the time. Same data, opposite conclusion; the offset is what makes the comparison fair.

## Common mistake

Adding exposure as an ordinary covariate ( $\beta$  estimated) “to be safe”. That lets the model bend the exposure effect away from proportionality and destroys the rate interpretation; the offset's whole point is  $\beta_{\text{exposure}} \equiv 1$ .